HPC Through the Ages

Is this at all interesting to an RSE?

All your questions answered by Simon and Simon

History

- HPC as we know it has been around for well over half a century
- So what has changed?   And why?
- The hardware has evolved, getting faster and larger memory
- Software is much longer lived than hardware and although it seems like it changes faster, much of the software we use has stuck around for a long time
- People? Well we are still here, but in smaller numbers compared to the increased compute power – how do we cope !!??

Hardware

- Here’s some history (show the chart)

Hardware outcomes

- With lots of detours and experiments en-route HPC hardware is now stable and well defined
- It is now all commodity! Processors, memory, interconnect – all started bespoke for HPC …
- But that does mean compromise eg:
    - Virtual memory
    - Cache (poor balance between memory and processor speeds)
    - Precision
    - Questionable energy efficiency
    - Limited interconnect performance

Software

- Here’s some history (show the chart)

Software outcomes

- Software has had to struggle constantly against changing hardware and vast perturbations in performance characteristics.
- Languages have evolved to meet our science needs and preserve the longevity of our code for DECADES!
- Fortran endures as being easy for scientists and engineers to express their maths. It has grown to support more modern programming methods and features.
- C evolved from the programming language for Unix into a general purpose programming language.
- C++ was created more recently to provide a richer language in more modern styles.

Software outcomes (2)

- As the hardware evolved from single faster processors to multi-node and multi-core to get more performance, the HPC software for using this hardware has kept pace, albeit with increased complexity.
- MPI provides the support for “long distance” communications between parallel tasks.
- OpenMP, which we thought might die a few years ago, is back with a vengeance to support a wider range of hardware.
- Combining the two into hybrid code allows us to address the largest problems.

Evolution of the profession

- Evolution of the profession itself:
  - I’ll do <thing>
  - the post-doc will do <thing>
  - someone else’s postdoc will do <thing>
  - …
  - RSE’s / RTP’s do <things>

Evolution of the software engineer

- Evolution of the person
  - Scientist
  - Scientific programmer
  - maintainer in their copious spare time
  - …
  - RSE

Evolution of the metaphors

- Evolution of the metaphors:
  - Unicorns
  - Third space
  - Glue / lube
  - Academic related
  - Not unicorns
  - Changelings / Chameleons
  - Trainers / explainers

Compromises

- The hardware outcomes mean we have to work round the limitations, which is where HPC software is unique
- It’s long lived (round of applause for Fortran!!!)
- It’s adaptable and evolutionary
- It’s shared
- It’s standard
- Keeps us on our toes!

Pat on the back everyone!

- Actually this is a pretty amazing achievement ☺
- We’ve evolved a vibrant software ecosystem that deals with changing and incompatible underlying hardware
- We’ve evolved too
- It works with a whole load of non-ideal (commodity) hardware
- It enables the research community to perform long running research programmes with considerable ease and breakthrough results with each new generation of systems
- Look back to the future!

End

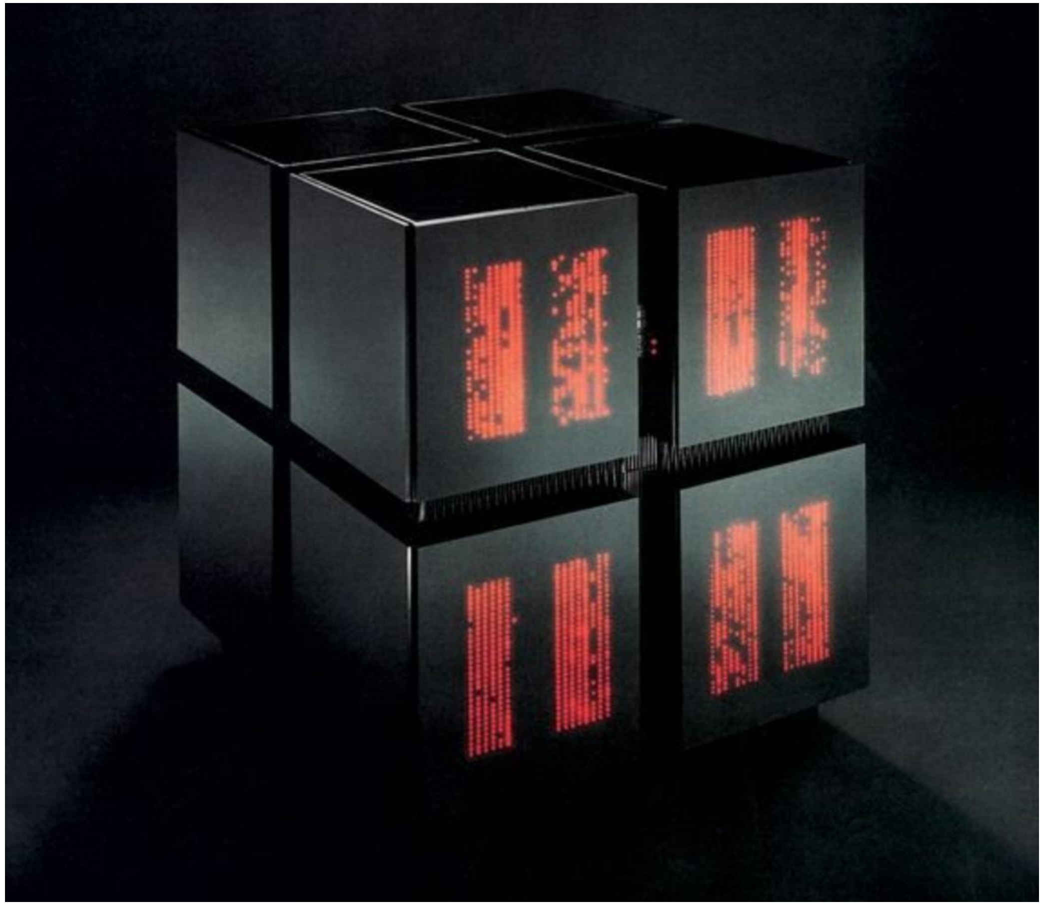
Bonus: historical quiz

From the pictures of 6 famous HPC systems:

- Identify the the computer
- Guess the special feature or capability of that machine

1

2



3

4

IBM's Blue Gene/L currently tops the table with 280 TFlops of computational grunt

5

6

Answers

1. beowulf cluster
    a. ethernet, linux, home build
2. Cray 1
    a. Vector processor, vector chaining, RISC
3. Connection Machine CM-1
    a. 1-bit processor, 3-d torus, dataflow architecture
4. Cray XMP
    a. Multi-processor, shared memory, scatter/gather, SSD
5. IBM Blue Gene
    a. balanced CPU/mem/interconnect architecture, RS6000 single chip, massively parallel distributed, Lattice QCD
6. SGI Origin 2000
    a. ccNUMA, hyper-cube, MIPS processors, massive mem 2TB, scales to 512 nodes